Implementation of a Basic Reinforcement Learning Model (Q-Learning)

Aim

To implement the **Q-Learning** algorithm, a model-free reinforcement learning technique in which an **agent** learns an optimal policy by interacting with an **environment** and updating a **Q-table** of state-action values, and to analyze its performance using **cumulative reward per episode, steps per episode, convergence of the Q-values** and the **learned policy**.

Algorithm

A. Q-Learning Implementation

Input: Environment (states **S**, actions **A**, rewards **R**); learning rate **α**; discount factor **γ**; exploration rate **ε**; number of episodes **N**

Step 1: Initialise the Q-table **Q(s, a) = 0** for all states s ∈ S and actions a ∈ A.

Step 2: For **each episode = 1 to N**:

1. Reset the environment and obtain the initial state **s**.
2. Repeat for each step until the **terminal state** (or maximum steps) is reached:
   - **Choose an action (ε-greedy policy):** with probability **ε** select a random action (**exploration**); otherwise select **a = argmax Q(s, a)** (**exploitation**).
   - **Take action a**, then observe the reward **r** and the next state **s′**.
   - **Update the Q-value (Bellman equation):**
     Q(s, a) ← Q(s, a) + α · [r + γ · max Q(s′, a′) − Q(s, a)]
   - **Move to the next state:** s ← s′.
3. **Decay ε** after each episode (for example, $\varepsilon \leftarrow \max(\varepsilon_{min}, \varepsilon \times \text{decay})$) to shift from exploration to exploitation.
4. Record the **total reward** and the **number of steps** of the episode.

Step 3: Output the learned Q-table and the optimal policy:
π(s) = argmax Q(s, a)

B. Performance Analysis Procedure

Step 1: Define the **environment** (for example, a Grid World or the FrozenLake environment) with its states, actions and rewards.

Step 2: Set the **hyperparameters** (α, γ, ε, ε decay, number of episodes).

Step 3: Train the agent using the Q-Learning algorithm.

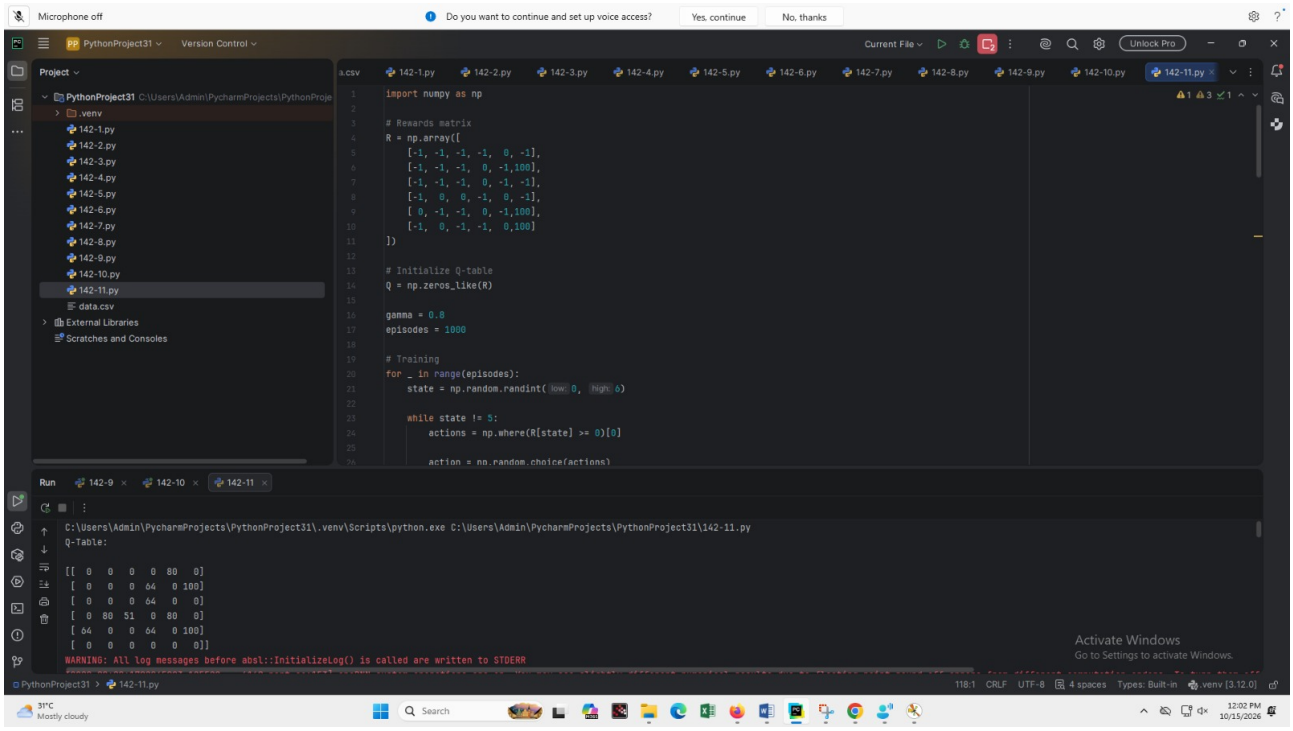
Step 4: Plot the **total reward per episode** and the **steps per episode** to observe learning progress.

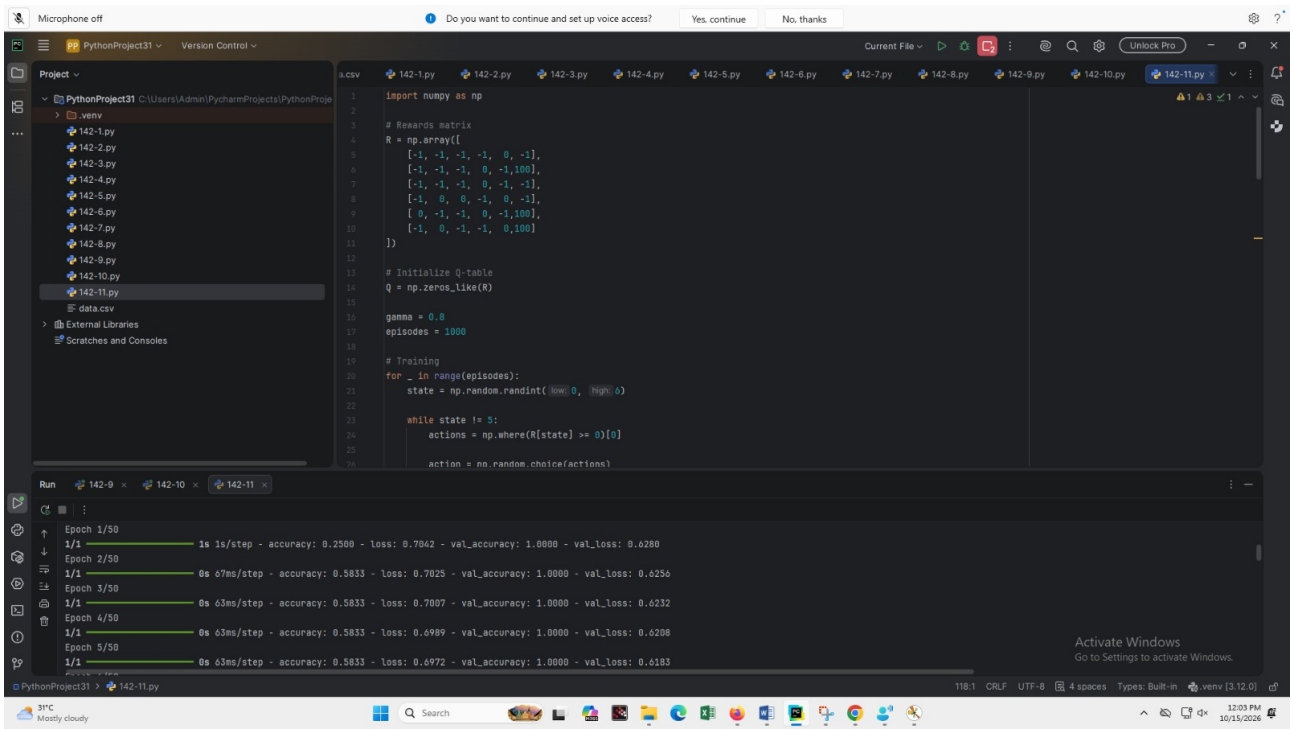
Step 5: Display the final **Q-table** and the **learned policy** (best action for each state).

Step 6: Test the trained agent with ε = 0 over several episodes and compute the **success rate** and **average reward**.

Step 7: Vary **α, γ and ε** and **compare** their effect on convergence speed and final performance, then **conclude**.

Program


import numpy as np
Rewards matrix
R = np.array([
[-1, -1, -1, -1, 0, -1],
[-1, -1, -1, 0, -1,100],
[-1, -1, -1, 0, -1, -1],
[-1, 0, 0, -1, 0, -1],
[0, -1, -1, 0, -1,100],
[-1, 0, -1, -1, 0,100]
])
Initialize Q-table
Q = np.zeros_like(R)
gamma = 0.8
episodes = 1000
Training
for _ in range(episodes):
state = np.random.randint(low: 0, high: 6)
while state != 5:
actions = np.where(R[state] >= 0)[0]
C:\Users\Admin\PycharmProjects\PythonProject31\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject31\142-11.py
Q-Table:
[[0 0 0 0 80 0]
[0 0 0 64 0 100]
[0 0 0 64 0 0]
[0 80 51 0 80 0]
[64 0 0 64 0 100]
[0 0 0 0 0 0]]
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
Activate Windows
Go to Settings to activate Windows.


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[-1, -1, -1, 0, -1,100],
[-1, -1, -1, 0, -1, -1],
[-1, 0, 0, -1, 0, -1],
[0, -1, -1, 0, -1,100],
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Epoch 1/50
1/1 1s 1s/step - accuracy: 0.2500 - loss: 0.7042 - val_accuracy: 1.0000 - val_loss: 0.6280
Epoch 2/50
1/1 0s 67ms/step - accuracy: 0.5833 - loss: 0.7025 - val_accuracy: 1.0000 - val_loss: 0.6256
Epoch 3/50
1/1 0s 63ms/step - accuracy: 0.5833 - loss: 0.7007 - val_accuracy: 1.0000 - val_loss: 0.6232
Epoch 4/50
1/1 0s 63ms/step - accuracy: 0.5833 - loss: 0.6989 - val_accuracy: 1.0000 - val_loss: 0.6208
Epoch 5/50
1/1 0s 63ms/step - accuracy: 0.5833 - loss: 0.6972 - val_accuracy: 1.0000 - val_loss: 0.6183
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